

Modeling human microbiology with AI

Outpost is building the AI layer for the largest unmodeled variable in human biology: the microbiome.

Outpost





Welcome, thanks for joining us!



Neythen Treloar, PhD

PRINCIPAL ML RESEARCHER

PhD in ML, mathematics,
and SynBio Foundation
models and RL.

AI for biological and
microbial data.

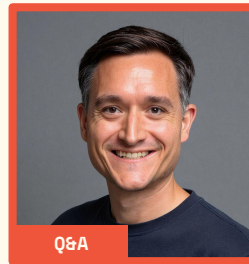


Saif Ur-Rehman, PhD

VP OF ENGINEERING

First engineering hire
at Basecamp Research
PhD in comp bio.

Scaled bioinformatics &
genomics infra.



Alex Merwin

COO, CO-FOUNDER

Worked at 2 unicorn
exited startups.

Ops leader at \$1B+ scale
at Amazon. Health & Bio
leadership at AWS.



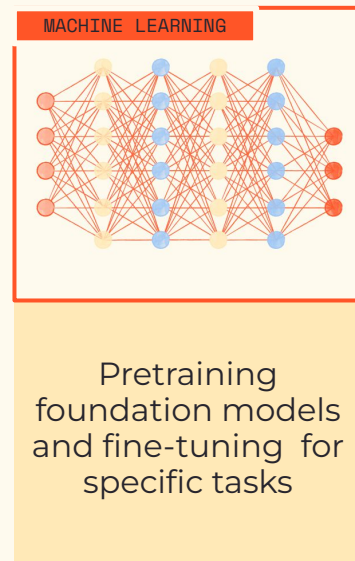
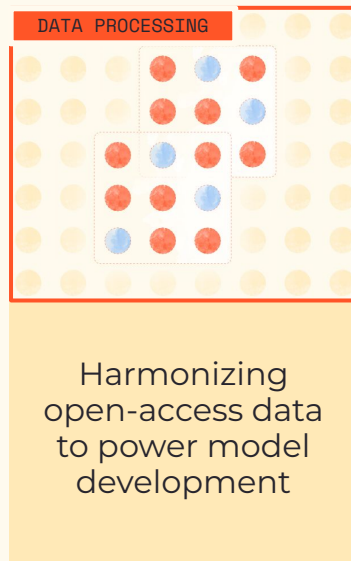
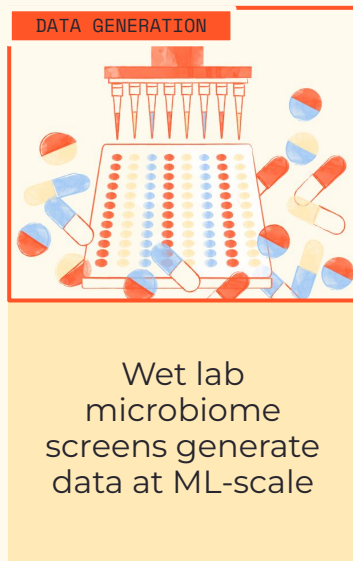
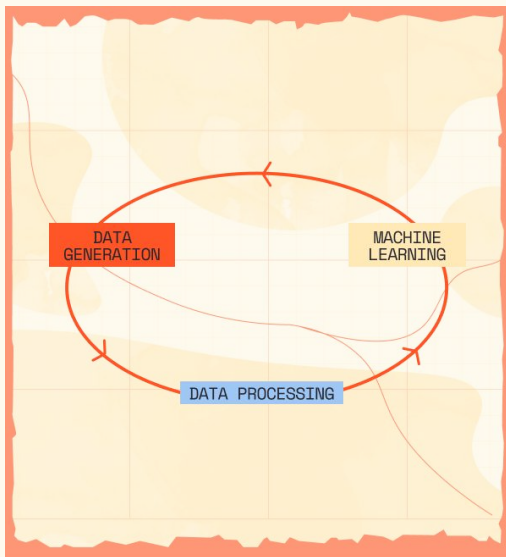
The microbiome has a macro impact on our health—but we don't understand it.

We coexist with trillions of microbes. They fundamentally shape how we experience health and disease.

Despite broad awareness, no one has cracked the code of how the microbiome drives health outcomes.

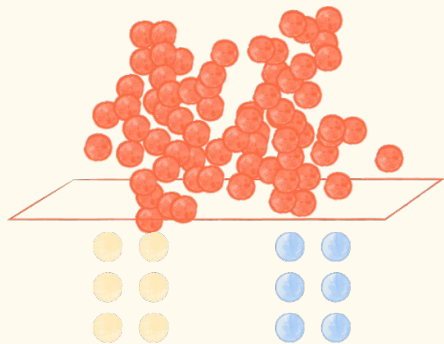


Outpost is building tools to make modeling the microbiome easier.





We're building foundation models for microbial communities

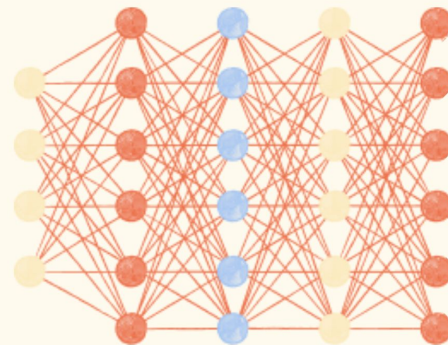


Trained on **diverse microbial community data** to learn the underlying biology

co-occurrence patterns, relative abundances, community structure, and ecological relationships between species.

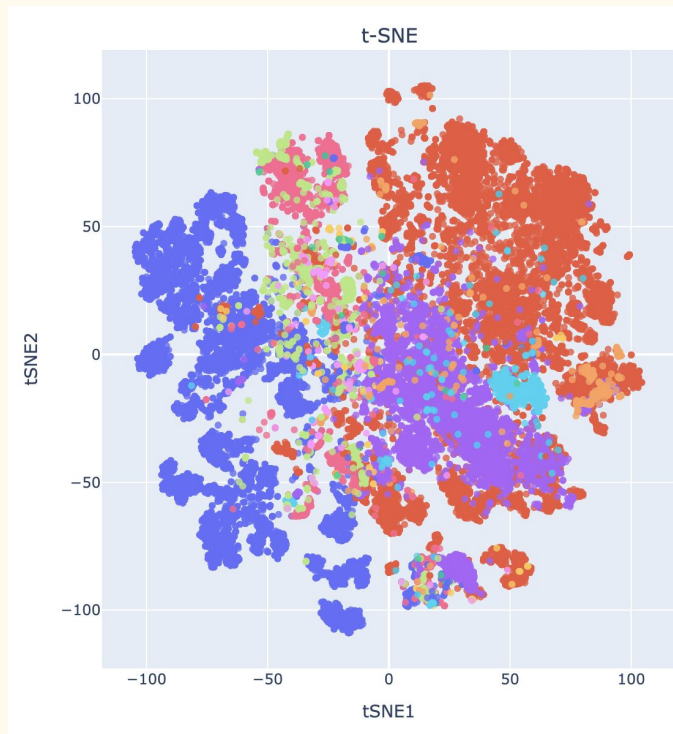
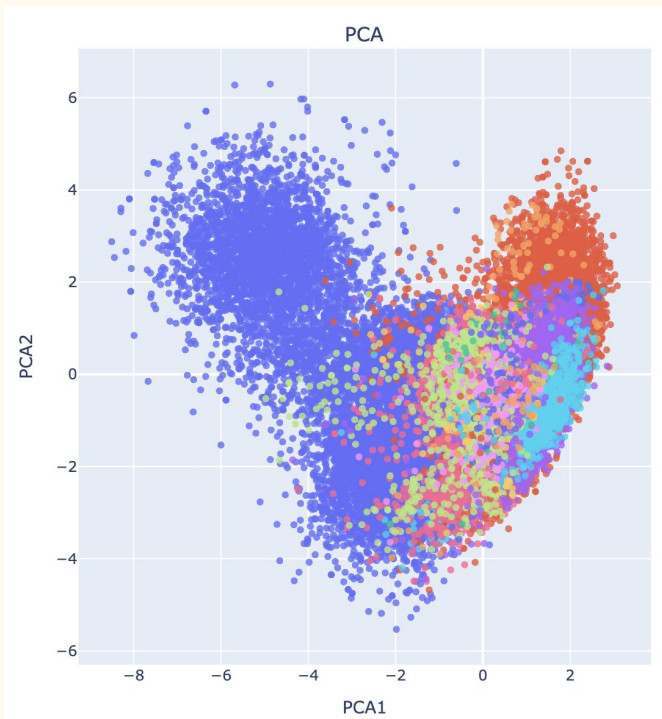
This model becomes the starting point for new modeling tasks:

- **Start from a strong baseline** instead of from scratch
- Get **better results with less data**
- **Use fewer resources** — the model already understands the fundamentals





Pretrained representations are meaningful



- Digestive System
- Marine
- Soil



Using a foundation model for your own applications

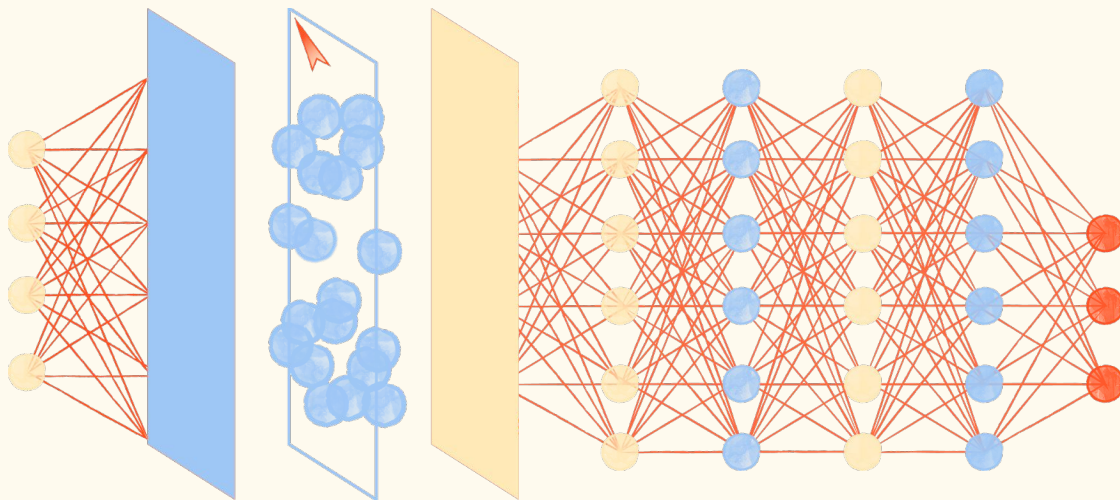
1 Waypoint, a foundation model of general microbial ecosystems

+

2 Small dataset for specific use case

=

3 Fit-for-purpose model for your prediction task



You don't need a massive dataset for every new question.

The foundation model brings biological context — your data teaches it the specifics.

The result is **better accuracy with less data** than training from scratch.



Using a foundation model for your own applications

1

Waypoint, a foundation model of general microbial ecosystems

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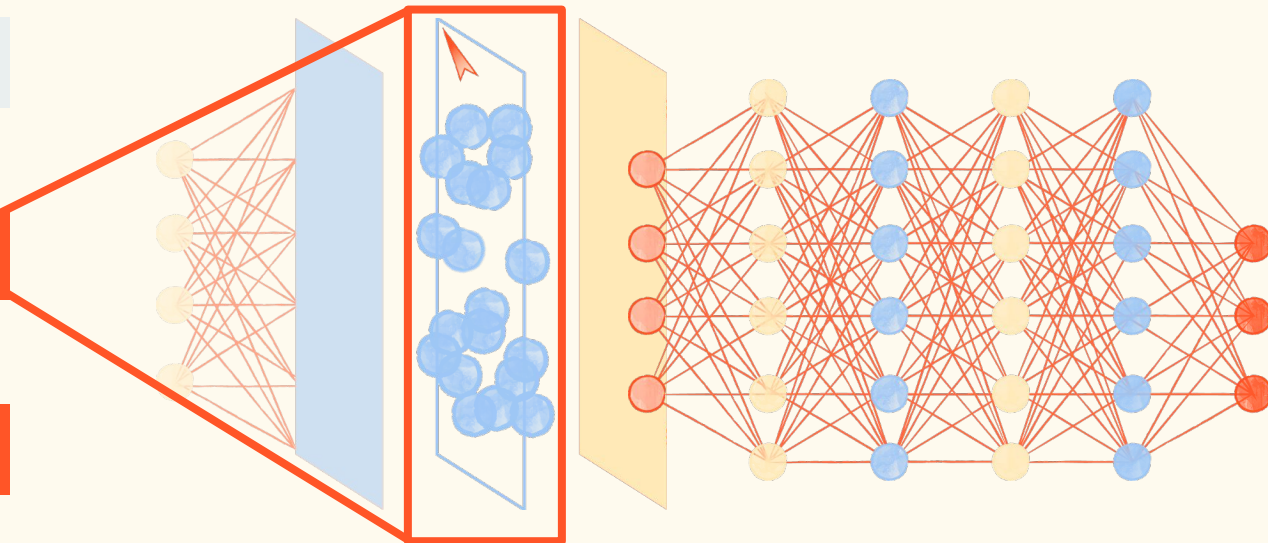
2

Small dataset of infant gut samples, paired with age

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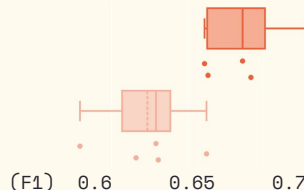
3

Fit-for-purpose model for infant age prediction



Model trained by fine-tuning infant samples on foundation model

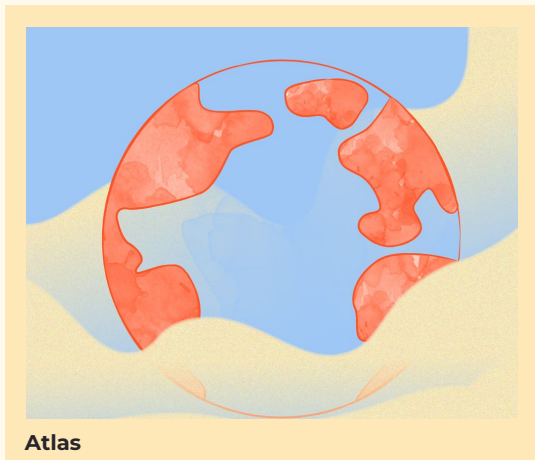
Models trained from scratch solely on infant gut samples



Using the foundation model as a starting point for training leads to **more accurate task-specific models.**



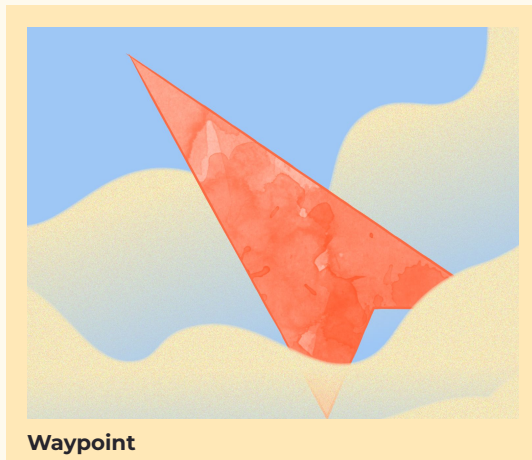
We open-sourced our model, dataset, and benchmarks



Atlas

TRAINING DATA

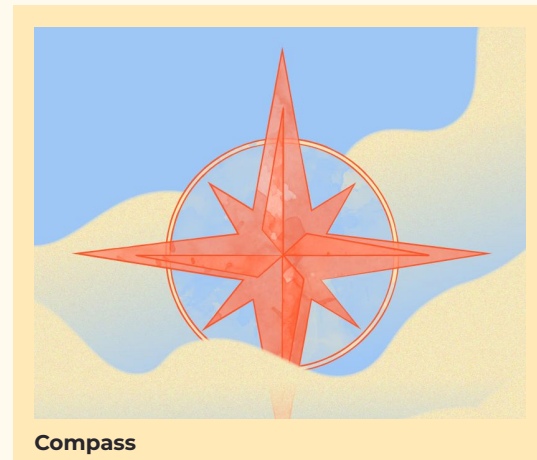
539, 308 Samples



Waypoint

FOUNDATION MODELS

3 PARAMETER SIZES



Compass

BENCHMARKS

8 PREDICTION TASKS



Hugging Face

<https://huggingface.co/outpost-bio>



GitHub

<https://github.com/Outpost-Bio/waypoint>



Tamarind Bio

<https://app.tamarind.bio/tools/waypoint>

bioRxiv
THE PREPRINT SERVER FOR BIOLOGY

<https://www.biorxiv.org/content/10.64898/2026.05.02.722381v1>





Plan

1. Embedding / analysis functionality on tamarind demo
2. Infant gut classification task notebook using the waypoint command line interface
3. Drug degradation regression task notebook using the waypoint command line interface



Tamarind Bio

Join the Waypoint Slack community

Outpost

